FreightOS Blueprint — Team 16 (Trans Ontario)

A human-first freight control tower: the dispatcher decides, the system checks, the help is optional.

1. The pitch and the principles

One-liner: “Dispatchers keep the wheel. The system keeps the checklist. The AI is a co-pilot you can switch off.”

Four principles (put these on your slide and in the app’s About page):

1. **Human decides, system enforces.** Everything can be done by hand. Hard rules block invalid moves; soft rules warn and allow an override with a logged reason.
2. **Help is a button, never a takeover.** Suggestions are proposals with a before/after diff and an Apply or Dismiss choice.
3. **Nothing is hidden.** Every field is badged Provided, Estimated or Missing. Every recommendation has a “Why?”.
4. **Everything is reversible.** Undo and an audit trail on every change.

2. What the real data tells you (use this, other teams will not)

From the Sample Driver Day workbook (11 weekdays, Sept 21 to Oct 3, 2026):

- About 250 stops and about 40 drivers per day, 71 trucks, 176 trailers, 1,285 customers. Mostly GTA (Toronto, Scarborough, Mississauga, Markham, Brampton, Vaughan).
- **Pickup AND delivery:** 1,118 pickups, 1,403 drop-offs. Order IDs come in pairs (e.g. `2986297P` / `2986297D`). The pickup must happen before its drop-off.
- **Service tiers:** LTL, OVERNIGHT, RUSH, RUSH90, SAME DAY, AFTER HOURS, OUT OF TOWN, FLAT BED, FTL.
- **Equipment:** truck, 53ft trailer, van, car, tractor and others. **Special handling:** crossdock (413 stops), dangerous goods (84), inside delivery, tailgate, pump truck, straps.
- **Weight is skewed:** median stop 650 lb, 173 stops over 10,000 lb.
- **Missing data:** appointments 94% blank, linear feet 69%, skid spots 65%, dim. weight 35%, cubic volume never recorded.

- **Dirty data:** 8% of stops are 0 minutes on site, some on-site times are absurd, and “late by” ranges from -804 to +3,176 minutes.
- **Opportunities:** on 110 of 907 pickup customer-days (12%), more than one driver visited the same customer. 54 of 159 appointment stops were late.
- **Not in the file:** truck capacities, coordinates, cubic volume. These must be documented assumptions.

The Sample Routing Report was not available when this was written. If you get it, map its columns to the data model in section 6.

3. How to stand out 10x

1. **Replay a real day and score yourself.** Run your planner on a real day’s orders and compare to what drivers actually did (km, driver-hours, late stops, trucks used). Put one number in the pitch.
2. **Pickup-and-delivery pairing** handled correctly, when most teams will treat it as plain delivery.
3. **Block vs warn with logged overrides**, like a real dispatch tool.
4. **Confidence badges** on every field, plus an in-app Assumptions page. This hits the “Data Transparency” requirement head-on.
5. **Ranked recovery options** for each disruption (cheapest, fastest, fewest changes) with a plan diff.
6. **Undo and audit trail**, showing every change as Manual or Suggested.
7. **Command bar (Ctrl+K)** so a dispatcher can jump anywhere or do anything in seconds.
8. **Built-in Demo Director** (section 8) so the demo is repeatable, controllable and never depends on luck.
9. **Plan resilience score**: how well the plan survives a breakdown, so slack is visible.
10. **Brand-native look:** it looks like it belongs to Trans Ontario.

4. Navigation: make it effortless

Layout (same on every screen):

- **Left sidebar (always visible):** Dashboard, Shipments, Fleet, Map, Exceptions (with a red count badge), Simulator, Impact, Assumptions.
- **Top bar:** search box (Ctrl+K command bar), simulated clock and date, Undo button, Demo Mode button, small AI assistant button.

- **Right-side detail drawer:** click any shipment, truck or exception and it opens in a drawer on the right. The dispatcher never loses their place in the table.

Rules that keep it fast:

- **Three-click rule:** any common task (assign, reassign, resolve an exception) takes at most three clicks.
- **Status chips everywhere** (Unassigned, Assigned, At risk, Late, Done), same colors on every screen.
- **Quick filters as one-click pills** above tables: Unassigned, At risk, Rush, Dangerous goods, Missing data.
- **Bulk select** with a floating action bar ("Assign 12 selected").
- **Quick assign** shows only compatible trucks, with the reason greyed-out trucks are excluded.
- **Every row has a primary action button** so nobody hunts through menus.
- **Inline editing** for weight, window, priority and notes.
- **Toasts with Undo** after every change.
- **Keyboard shortcuts:** `/` search, `U` unassigned filter, `E` exceptions, `Z` undo, `?` shortcut cheat sheet.
- **Dashboard cards are links:** clicking "14 at risk" opens Shipments already filtered to those.

5. The five layers of help (manual first)

Layer 0: fully manual. View, search, filter, edit, assign (dropdown, drag-and-drop, bulk), reorder stops, change priority, mark failed delivery, take a truck out of service.

Layer 1: always-on rules (no AI). Run on every action.

Rule	Type	Behavior
Weight over truck capacity	Hard block	Refuse, explain
Linear feet or skid spots over capacity	Hard block	Refuse, explain
Equipment mismatch (e.g. 53ft requested, van chosen)	Hard block	Refuse, explain
Dangerous goods on uncertified truck or driver	Hard block	Refuse, explain

Rule	Type	Behavior
Driver hours would be exceeded	Hard block	Refuse, explain
Delivery sequenced before its pickup	Hard block	Refuse, explain
Appointment or deadline cannot be met	Soft warning	Allow with logged reason
Rush or same-day tier at risk	Soft warning	Allow with logged reason
Missing or estimated data used in a decision	Soft warning	Show badge and note
Heavy stop loaded in an awkward position	Soft warning	Show note

Verify the exact Canadian hours-of-service limits before hardcoding them (commonly cited as 13 hours driving and 14 hours on duty per day); label them as an assumption in the app.

Layer 2: on-demand suggestions (buttons).

- “Suggest trucks” for selected shipments.
- “Suggest full plan” for the day.
- “Fix this” on any exception, returning 2-3 ranked options with a diff.
- “Find merge candidates” for the consolidation finder.

Layer 3: optional AI assistant (corner button). Explains a warning in plain English, compares options, answers “what happens if truck 12 is out?”. It can only propose; the dispatcher applies.

Layer 4: predictions (optional, labelled as estimates). Expected dwell time range per stop and a late-risk score per delivery.

6. Data model

Order (one per P or D leg): order_id, pair_id, leg (pickup or delivery), customer_code, address, lat, lng, service_tier, equipment_requested, special_handling, pieces, piece_type, weight_lb, linear_feet, skid_spots, appointment_start, appointment_end, status, assigned_vehicle_id, stop_sequence, plus a **confidence** flag per key field (provided, estimated, missing) and a source note.

Vehicle: vehicle_id, type (van, straight truck, tractor + 53ft trailer, flat bed, car), max_weight_lb, max_linear_ft, max_skids, equipment_tags, dangerous_goods_certified, status (active, down).

Driver: driver_id, name, licence_class, dangerous_goods_certified, hours_driven_today, hours_on_duty_today, status (available, sick, off).

Plan: version_id, created_by (manual or suggested), vehicle_id, ordered stop list, planned arrival times, risk flags.

Event log: timestamp, actor (dispatcher, system, suggestion), action, before, after, reason. Powers Undo, audit trail and the replay.

Assumption table (capacities are placeholders, document and adjust):

Vehicle type	Weight (lb)	Linear ft	Skid spots
Cargo van	3,000	10	3
Straight truck	10,000	24	10
Tractor + 53ft trailer	43,000	53	26
Flat bed	40,000	48	24

Where linear feet or skids are missing, estimate from pieces, piece type and weight, and badge the field **Estimated**.

7. The suggestion engine

This is the part to be honest about in a no-code builder: a full solver library may not be available. Pick one:

- **If your builder can run Python (Antigravity can):** use Google OR-Tools (pickup-and-delivery with time windows and capacities).
- **If not (or to stay safe):** implement a simple heuristic in JavaScript: sort by service tier and window, insert each order into the best compatible route (lowest added travel time), then try swaps between routes for improvement. Reject any insertion the rules engine blocks.

Travel time: use straight-line distance times a city factor, or an API if available. The data has real GPS travel minutes between stops, which you can use to calibrate the factor.

Recovery options for a disruption:

1. Collect the affected, not-yet-completed orders.
2. Try reassigning to other trucks (rules engine checks each).
3. Try handing off to a partner carrier at an estimated cost.
4. Score each option on cost, lateness and number of changes. Show the best 3: **Cheapest, Fastest, Fewest changes**, each with a plain-English “Why?”.

Plan diff view: orders moved (from truck to truck), orders sent to partner, new at-risk deliveries, change in km, hours and cost.

8. Built-in Demo Director (the quick demo, inside the app)

A **Demo Mode** button in the top bar loads a pre-chosen real day and opens a small stepper panel. The demo is scripted, repeatable and fully controllable, so a live glitch cannot ruin it.

Controls: Next, Back, Auto-play, Reset to start, Speed (1x, 10x, 60x), “Jump to step” menu, and a Presenter toggle that hides narration for judges who want to explore. Include a **Try it yourself** mode that leaves everything unlocked.

Scripted steps (about 5 minutes):

1. **8:00 Morning snapshot.** Dashboard shows about 240 orders, several unassigned, 3 flagged problems. Narration: “This is a real day’s work.”
2. **Blocked assignment.** Auto-select a dangerous-goods shipment and try a non-certified truck. The system blocks it and explains why. Then show a warning-only case with an override reason.
3. **Suggest full plan.** Click once. Plan appears on the map; Impact panel shows the baseline vs. plan.
4. **10:15 Breakdown.** A truck goes down. Exceptions inbox lights up, affected shipments highlight. Click “Fix this”, review 3 ranked options with the diff, apply one.
5. **11:30 Late rush order.** Show insert-or-partner decision with cost.
6. **14:00 Failed delivery.** Offer reschedule and show impact on that truck’s remaining stops.
7. **End of day Impact.** Real day vs. your plan: km, driver-hours, late stops, trucks used, resilience score.

How to build it: store the script as a JSON list of steps, each with a clock time, an action (load day, attempt assignment, run plan, trigger disruption, add order), a target id, and a narration card. Use a fixed seed so dwell-time samples are identical every run. Pre-run each step once and save the results as fallback data so the demo works even if the live computation is slow.

Backup: record a 3-minute screen capture of the full demo and keep it on the presenter's laptop.

9. Visual design (Trans Ontario brand)

- **Colors:** Royal Blue `#1F4AEE` (primary actions, links), Black Russian `#030718` (sidebar, headers, text), White Smoke `#F2F2F2` (page background), White `#FFFFFF` (cards).
- **Font:** Montserrat (Light, Regular, Bold).
- **Logo:** use the supplied all-blue logo top-left of the sidebar.
- **Alert colors only for alerts:** red (blocked or late), amber (warning or at risk), green (on track). Keep them out of normal UI so alerts stand out.
- **Style:** generous spacing, large readable numbers on the dashboard, rounded cards, subtle shadows, no clutter.

10. Build plan for your AI builder (Base44 / Antigravity)

AI builders work best with **one focused prompt per phase**. Finish and test each phase before sending the next. Upload the cleaned data as a CSV or JSON when prompted.

Phase 0: Setup and shell

> Build a web app called FreightOS for a freight carrier called Trans Ontario. Left sidebar (always visible) with: Dashboard, Shipments, Fleet, Map, Exceptions (with a red count badge), Simulator, Impact, Assumptions. Top bar with search, simulated clock, Undo button, Demo Mode button and a small assistant button. Clicking any row opens a detail drawer on the right. Use the colors Royal Blue #1F4AEE, Black Russian #030718, White Smoke #F2F2F2, White, and the Montserrat font. Clean, spacious, professional.

Phase 1: Data

> Create entities for Order, Vehicle, Driver, Plan and EventLog with these fields: [paste the data model from section 6]. Import this orders file [upload]. Every key field has a confidence flag (provided, estimated, missing) shown as a small badge.

Phase 2: Shipments and Fleet (manual control)

> Shipments page: searchable, sortable table with quick-filter pills (Unassigned, At risk, Rush, Dangerous goods, Missing data), bulk select with a floating action bar, inline editing, and a Quick Assign button that lists only compatible vehicles. Fleet page: vehicles with driver, capacity used (weight, linear ft, skids), hours used and an expandable stop list. Reassign via dropdown or drag and drop.

Phase 3: Rules engine and Undo

> Add a validation function that runs on every assignment. Hard blocks (refuse and explain): [paste hard-block rows from section 5]. Soft warnings (allow with a required reason that is logged): [paste soft rows]. Log every change in EventLog with actor, before, after and reason. The top-bar Undo button reverses the last change. Show a toast with Undo after each action.

Phase 4: Exceptions and manual breakdown

> Exceptions page: one inbox of issues (breakdowns, overloads, missing info, at-risk deliveries), each with a reason and an action button. Add a Mark Truck Down button on a vehicle. When pressed, highlight all affected incomplete orders and show compatible replacement trucks so the dispatcher can reassign manually.

Phase 5: Suggestions

> Add Suggest Trucks (selected orders), Suggest Full Plan, and Fix This on exceptions. Use a heuristic planner: sort by service tier and time window, insert each order into the compatible route with the lowest added travel time, then try swaps. Pickup must come before its paired delivery. Every suggestion is shown as a proposal with a before/after diff and Apply or Dismiss. For disruptions show three options: Cheapest, Fastest, Fewest changes, each with a Why explanation. Include a partner-carrier handoff option with an estimated cost.

Phase 6: Simulator and Demo Director

> Simulator page with a clock (play, pause, speed 1x/10x/60x) that advances the day and fires scheduled events. Add a Demo Mode button that opens a step panel (Next, Back, Auto-play, Reset, Jump to step, Presenter toggle) driven by a JSON script of steps: [paste the seven steps from section 8]. Use a fixed random seed so every run is identical.

Phase 7: Impact and Assumptions

> Impact page comparing the planned day to the real day from the data: total km, driver-hours, trucks used, late stops, utilization, and a plan resilience score. Assumptions page listing every simplification, estimated field and capacity placeholder in plain language.

Phase 8: Polish (only if time)

> Add the Ctrl+K command bar, keyboard shortcuts, a truck load diagram (stops in unload order), dwell-time range estimates, merge-candidate finder, and the optional AI assistant that explains warnings and compares options but never applies changes itself.

Tips for the builder:

- Ask it to **list what it built after each phase** so you can check against this blueprint.
- When something breaks, give it the exact error or a screenshot and say what you expected.
- Don’t add features mid-phase. Finish, test, then move on.
- Keep a copy of the working app before each big change.

11. Assumptions to document (they are explicitly allowed)

- Vehicle capacities are placeholders by vehicle type.
- Missing linear feet and skid spots are estimated from pieces, piece type and weight.
- Coordinates come from geocoding customer addresses once and caching them.
- Travel times are approximations calibrated against GPS data in the file.
- Hours-of-service limits are simplified; confirm exact rules.

- Dwell times are estimated from historical medians, with outliers removed.
- Partner-carrier costs are illustrative.

12. Pitch outline (5 minutes)

1. **The problem** in one sentence, using the challenge statement.
2. **Our philosophy:** human decides, system enforces, help optional.
3. **Live Demo Director run** (section 8).
4. **The number:** “On a real day, versus what actually happened, we’d save X km and Y driver-hours, and catch Z problems before dispatch.”
5. **Trust:** show a confidence badge, a “Why?”, and an override log.
6. **Next steps:** real TMS integration, live GPS feed, learned travel and dwell models.

13. Risks and fallbacks

- **Builder can’t run a solver:** use the heuristic planner (section 7).
- **Geocoding takes time:** do it first and cache.
- **Scope creep:** follow the phases strictly; Phases 0 to 4 alone are a complete manual system.
- **Demo glitch:** pre-run steps, saved fallback data, recorded backup video.
- **Data dirty:** document your cleaning rules in the Assumptions page.

14. Final checklist before judging

- [ ] Demo Mode runs start to finish from a fresh reset.
- [ ] At least one blocked assignment, one override, one undo shown.
- [ ] Breakdown shows 3 ranked options and a diff.
- [ ] Impact page has real numbers vs. the real day.
- [ ] Assumptions page is complete.
- [ ] Brand colors, font and logo applied.
- [ ] Backup video recorded.
- [ ] Everyone can explain one screen and one design decision.